

Debdeep Bhattacharya

CONTACT INFORMATION

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RESEARCH INTERESTS

I work on mathematics of materials, in particular, on modeling fracture propagation in solids, and determining effective or homogenized properties of brittle granular media. I use high-performance computing and analysis of partial differential equations, especially of nonlocal and nonlinear types to solve emerging problems in the intersection of mathematics and material science. I am also interested in applying signal processing and machine learning techniques in real-world problems.

EMPLOYMENT

Grinnell College

Assistant Professor, Department of Mathematics, 2024 - present

University of Utah

Postdoctoral Researcher, Department of Mathematics, 2023 - 2024

Mentor: Ken Golden

Louisiana State University

Postdoctoral Researcher, Department of Mathematics, 2020 - 2023

Mentor: Robert Lipton

EDUCATION

The George Washington University

Ph.D. in Mathematics, 2020

Advisor: Frank Baginski

Thesis title: Harmonic Analysis Techniques in Nonlinear Dispersive Equations and Signal Processing

Tata Institute of Fundamental Research Centre for Applicable Mathematics, Bengaluru, India

Master of Science (MSc) in Mathematics, May 2014

Indian Statistical Institute, Bengaluru, India

Bachelor in Mathematics, May 2012

RESEARCH VISITS

Oak Ridge National Laboratory

Mathematical Sciences Graduate Internship (MSGI), National Science Foundation (NSF), Summer 2019

Supervisor: Dr. Pablo Seleson

University of Hawai'i at Manoa

Visiting Scholar, Department of Physics & Astronomy, Summer 2018

Supervisor: Peter Gorham

PUBLICATIONS AND PREPRINTS

1. Debdeep Bhattacharya, Davood Damircheli, and Robert P Lipton. Crushing, comminution and fracture: Extreme particle deformation in three-dimensional granular aggregates. *arXiv preprint arXiv:2506.05362*, 2025
2. Debdeep Bhattacharya, Tyler P Evans, and Andrej Cherkaev. Design of resilient structures by randomization and bistability. *International Journal of Engineering Science*, 215:104296, 2025
3. Debdeep Bhattacharya and Robert Lipton. Macroscopic effects of intraparticle fracture, grain topology and shape on vehicle dynamics and mobility over gravel road beds. *Granular Matter*, 27(1):2, 2025
4. Robert P. Lipton and Debdeep Bhattacharya. Energy balance and damage for dynamic fast crack growth from a nonlocal formulation. *Journal of Elasticity*, 157(1):5, Nov 2024
5. Debdeep Bhattacharya, Robert Lipton, and Patrick Diehl. Quasistatic fracture evolution using a nonlocal cohesive model. *International Journal of Fracture*, pages 1–18, 2023
6. Debdeep Bhattacharya and Robert P. Lipton. Simulating grain shape effects and damage in granular media using PeriDEM. *SIAM Journal on Scientific Computing*, 45(1):B1–B26, 2023
7. Prashant K Jha, Prathamesh S Desai, Debdeep Bhattacharya, and Robert Lipton. Peridynamics-based discrete element method (PeriDEM) model of granular systems involving breakage of arbitrarily shaped particles. *Journal of the Mechanics and Physics of Solids*, 151:104376, 2021
8. Debdeep Bhattacharya, Patrick Diehl, and Robert P Lipton. Peridynamics for quasistatic fracture modeling. In *ASME International Mechanical Engineering Congress and Exposition*, volume 85680, page V012T12A041. American Society of Mechanical Engineers, 2021
9. Debdeep Bhattacharya and Robert P Lipton. Quasistatic evolution with unstable forces. *Multiscale Modeling & Simulation*, 21(2):598–623, 2023
10. Debdeep Bhattacharya. Mass-concentration of low-regularity blow-up solutions to the focusing 2D modified Zakharov–Kuznetsov equation. *Partial Differential Equations and Applications*, 2(6):1–29, 2021
11. Debdeep Bhattacharya, Luiz Gustavo Farah, and Svetlana Roudenko. Global well-posedness for low regularity data in the 2D modified Zakharov-Kuznetsov equation. *Journal of Differential Equations*, 268(12):7962–7997, 2020
12. PW Gorham, A Ludwig, C Deaconu, P Cao, P Allison, O Banerjee, L Batten, D Bhattacharya, JJ Beatty, K Belov, et al. Unusual near-horizon cosmic-ray-like events observed by ANITA-IV. *Physical review letters*, 126(7):071103, 2021
13. Frank Baginski and Debdeep Bhattacharya. Modeling the launch dynamics of a large scientific balloon. *43rd COSPAR Scientific Assembly. Held 28 January-4 February*, 43:2311, 2021
14. Debdeep Bhattacharya. *Harmonic Analysis Techniques in Nonlinear Dispersive Equations and Signal Processing*. PhD thesis, The George Washington University, 2020

TECHNICAL REPORTS

1. Deconvolution problem and application to ANITA signals. Submitted to ANITA collaboration at University of Hawai'i at Manoa ([link](#))
2. Reduction of three-dimensional axisymmetric problems to two dimensions in Peridynamics. Submitted to the NSF as part of MSGI program ([link](#))

TEACHING
EXPERIENCE

As Instructor

- **Department of Mathematics, Grinnell College**
 - Fall 2025: Numerical Analysis (MAT 313)
 - Fall 2025: Calculus I (MAT 313)
 - Spring 2025: Linear Algebra (MAT 215)
 - Spring 2025: Differential Equations (MAT 220)
 - Fall 2024: Linear Algebra (MAT 215)
 - Fall 2024: Numerical Analysis (MAT 313)
- **Department of Mathematics, University of Utah**
 - Fall 2023: Partial Differential Equations (Math 3150)
 - Fall 2023: Business Algebra (Math 1090)
- **Department of Mathematics, Louisiana State University**
 - Fall 2022: Mathematical Methods in Engineering (Math 2070)
- **Department of Mathematics, George Washington University**
 - Summer 2017: Linear Algebra I (MATH 2184)
 - Summer 2016: Calculus with Pre-calculus I (MATH 1220)

As Teaching Assistant

- Department of Mathematics, George Washington University
 - Spring 2020: Calculus for the Social and Management Sciences
 - Fall 2019: Calculus III
 - Spring 2019: Calculus I
 - Fall 2018: Calculus II
 - Spring 2018: Calculus I
 - Fall 2017: Calculus with Pre-calculus I
 - Spring 2017: Partial Differential Equation
 - Fall 2016: Calculus I
 - Spring 2016: Calculus for the Social and Management Sciences
 - Fall 2015: Calculus with Pre-calculus I
 - 2016 - 2020: Calculus Lab

MENTORING

- Hung Chu and Regan Sambaugh (undergraduate students at Grinnell College) for a Mentored Advanced Project (MAP) on structural resilience of bistable lattices (Summer 2025)
- Kevin Zheng and Yuhan (Rain) Yan (undergraduate students at Grinnell College) for a Mentored Advanced Project (MAP) on the mechanical entanglement of granular aggregates of nonconvex shapes (Summer 2025)
- Nuwanthi Samarawickrama (graduate student at Louisiana State University) on a research project on *Well-posedness of state-based quasistatic evolution* (2022 January - 2023 May)
- Rigoberto Moncada Lopez (graduate student at California Institute of Technology) on a research project on *Hopper flow simulation of deformable grains using peridynamics* (2022 January - 2023 May)

PROFESSIONAL DEVELOPMENT	• I am a 2025 Project NExT fellow. • <i>Math teacher training workshop</i>, a week-long training event hosted the University of Utah, Fall 2023. • Fall 2022 cohort of <i>Evidence-Based Undergraduate STEM teaching</i> course by the Integration of Research, Teaching and Learning (CIRTL) network that prepares early career faculties in inclusive and evidence-based teaching methodologies.
HONORS AND ACHIEVEMENTS	• Received research allocation of 500,000 SU (Service Units) from the High Performance Computing Resource Allocation Committee, Louisiana State University, as a PI for project titled <i>Perigrain - granular media simulations with peridynamics</i>, 2022 January - December • SIAM award for outstanding efforts and accomplishments as a graduate student, 2018 • Dean's Graduate Conference Travel Grant, The George Washington University, 2017 • Columbian College of Arts of Sciences Fellowship, The George Washington University, 2015 – 2020 • Junior Research Fellowship from Tata Institute of Fundamental Research, India, 2012–2014 • INSPIRE Scholarship from Department of Science and Technology, Government of India, 2010-2012 • Student Fellowship from Indian Statistical Institute, 2009-2012 • Secured an all-India rank 31 in National Eligibility Test (NET) jointly conducted by Council of Scientific and Industrial Research and University Grant Commission, Government of India, 2013
ORGANIZED SEMINARS AND SYMPOSIA	• Minisymposium on <i>Nonlocal models continuum mechanics and granular media</i> at the 5th Annual Meeting of the SIAM Texas-Louisiana Section, University of Houston, November 4-6, 2022 • Minisymposium on <i>Theoretical and computational aspects of nonlocal operators</i> at the 7th SIAM Central States Annual Meeting, Oklahoma State University, October 1-2, 2022 • Minisymposium on <i>Peridynamic Theory and Multiscale Methods for Complex Material Behavior</i> at the 9th GACM Colloquium on Computational Mechanics for Young Scientists from Academia and Industry in Essen, Germany, September 21-23, 2022 • Spring 2019 Conference on Applied Mathematics, George Washington University, Washington, D.C., May 4, 2019 • GWU-SIAM conference on Applied Mathematics, George Washington University, Washington, D.C., April 29, 2017
REVIEWER FOR JOURNALS	• Engineering Fracture Mechanics • Journal of Peridynamics and nonlocal modeling • Journal of Open Source Software • Computational Geoscience
OUTREACH	• Volunteered at the National Math Festival in the SIAM booth to demonstrate mathematical concepts to high-school level students, Washington, D.C., 2017, 2018 • Served as the vice president of the SIAM chapter at the George Washington University, January 2016 - 2018 • Served as a faculty mentor at the SIAM TX-LA Graduate Mentoring Program, University of Houston, Texas, October 2022 (Mentee: Lander Besabe)

PROGRAMMING
SKILL

Languages: C/C++, Python, R, MATLAB

- High performance computing: MPI, openmp, CUDA, CuPy, mpi4py
- Machine learning: PyTorch, Tensorflow, scikit-learn, Keras
- Data manipulation: Pandas, SQL, Excel
- Visualization: matplotlib, ggplot2, seaborn, VisPy, VisIt, gmsh, meshio
- Other tools: BASH, L^AT_EX, Git, Vim

SOFTWARE
AUTHORED

GitHub profile: <https://github.com/debdeepbh>

- **perigrain:** A high-fidelity granular media simulation and analysis platform with deformation and fracture in arbitrary grain shapes using peridynamics and discrete element method. Applications include vehicle mobility on gravels of different shapes, hopper flow of deformable particles, among others.
- **Balloon launch dynamics:** 3D simulation and analysis of the launch dynamics of high altitude balloons in python with MPI Parallelization
- **libWTools:** Signal processing library using Fourier and wavelet-based tools
- **deconvolution:** Recovering electromagnetic signals from blurred and noisy observations from array antennas using Fourier and wavelet analysis techniques, and by solving regularized optimization problems
- **crack:** Numerical simulations of crack branching in sodalime glass using peridynamics
- **rexpense:** Generating complex expense reports and statistics in a multi-user setup
- **ml:** Machine learning projects with mathematical explanations
- **mathabotface:** Automated theorem and lemma twitting bot

INVITED TALKS

- 18th U.S. National Congress on Computational Mechanics, Chicago, IL, July 20 - 24, 2025
- Engineering Mechanics Institute Conference (EMI 2025), Anaheim, CA, May 27 - 30, 2025
- SIAM Conference on Mathematical Aspects of Materials Science (MS24), Pittsburgh, Pennsylvania, USA, May 19 - 23, 2024
- Department of Mathematics Seminar, University of Rutgers at Camden, Camden, New Jersey, USA, February 9, 2024
- Applied Math Seminar, San Diego State University, San Diego, California, USA, February 12, 2024
- Advances in Computational Mechanics (ACM 2023), University of Texas at Austin, Austin, Texas, USA, October 22, 2023
- Applied Math Seminar, University of Alabama, Huntsville, Alabama, USA, February 16, 2023
- Workshop on Scale-Bridging Materials Modeling at Extreme Computational Scales, University of California, Los Angeles, USA, April 18, 2023
- 5th Annual Meeting of the SIAM Texas-Louisiana Section, University of Houston, Texas, USA, November 5, 2022
- 19th U.S. National Congress on Theoretical and Applied Mechanics, University of Texas at Austin, Austin, USA, June 19, 2022
- Society of Engineering Science Annual Technical Meeting, Texas A&M University, College Station, Texas, April 26, 2022
- University of Nebraska-Lincoln, February 24, 2022, Lincoln, USA
- Applied Math Seminar, George Washington University, October 1, 2021, Washington, D.C., USA
- 16th U.S. National Congress on Computational Mechanics (USCNCM16), July 25, 2021, Chicago, USA (Virtual)
- ALOP Workshop on Nonlocal Models, July 13, 2021, Universität Trier, Germany

- (Virtual)
- MURI seminar series, California Institute of Technology, March 16, 2021, Pasadena, CA (Virtual)
 - 3rd Annual Meeting of the SIAM Texas-Louisiana Section, October 16-18, 2020, College Station, TX, USA (Virtual)
 - MURI seminar series, California Institute of Technology, September 9, 2020, Pasadena, CA (Virtual)
 - 13th AIMS Conference on Dynamical Systems, Differential Equations and Applications, June 5-9, 2020, Atlanta, USA
 - Computational and Applied Math (CAM) seminar, Computer Science and Mathematics Division, Oak Ridge National Laboratory, August 8, 2019
 - RIT in Applied Harmonic Analysis, Norbert Weiner Center, University of Maryland, May 13, 2019
 - Spring 2019 conference on Applied Mathematics, George Washington University, May 4, 2019
 - Antarctic Impulse Transient Anetann (ANITA) collaboration, December 10, 2018
 - Department of Physics, University of Hawai'i at Manoa, June 28, 2018
 - Graduate Student Seminar, The George Washington University, October 27, 2017
- POSTER PRESENTATIONS
- Workshop on Experimental and Computational Fracture Mechanics: Validating peridynamics and phase field models for fracture prediction and experimental design, Louisiana State University, Baton Rouge, Louisiana, USA, February 26-28, 2020
 - Oak Ridge Postdoctoral Association (ORPA) Research Symposium, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA, August 6, 2019
 - 2019 Workshop on Nonlinear Dispersive Partial Differential Equations and Inverse Scattering, The Fields Institute, Toronto, Canada, May 21 - 24, 2019
 - GW Research Days, George Washington University, Washington, D.C., USA, April 9, 2019
- CONFERENCE AND WORKSHOP PARTICIPATION
- Workshop on *Experimental and Computational Fracture Mechanics: Validating peridynamics and phase field models for fracture prediction and experimental design*, February 26-28, 2020, Baton Rouge, Louisiana, USA
 - 2019 *Workshop on Nonlinear Dispersive Partial Differential Equations and Inverse Scattering*, The Fields Institute, Toronto, Canada, May 21 - 24, 2019
 - *February Fourier Talks*, February 21-22, 2019, University of Maryland, USA
 - *IAS/PCMI 2018 Summer Graduate School on Harmonic Analysis*, July 1-21, 2018, Park City, Utah, USA
 - *February Fourier Talks*, February 15-16, 2018, University of Maryland, USA
 - Workshop on *Dispersive Equations, Solitons, and Blow-up*, September 4-8, 2017, Hausdorff Center of Mathematics, Bonn, Germany
 - *French-American Conference on Nonlinear Dispersive PDEs*, June 12-16, 2017, Centre International de Rencontres Mathématiques (CIRM), Luminy, Marseille, France
 - Research School on *Random Structures in Statistical Mechanics and Mathematical Physics*, March 6 -10, 2017, Centre International de Rencontres Mathématiques (CIRM), Luminy, Marseille, France
 - PDE/Analysis Mini School on *Dynamics of the energy critical wave equations* by Thomas Duyckaerts, University of North Carolina, Chapel Hill, 13-15 February, 2017
 - PDE/Analysis Mini School on *Random Schrödinger operators: Basic properties, localization, and spectral statistics* by Peter Hislop, University of North Carolina, Chapel Hill, 27-28 October 2016

- Workshop on *Getting Started with PDEs*, The Hebrew University, Jerusalem, Israel, September 11 - September 15, 2016
- *Third Chicago Summer School In Analysis*, University of Chicago, June 13 - June 24, 2016
- PIRE-CNA 2016 Summer School on *New Frontiers in Nonlinear Analysis for Materials*, Carnegie Mellon University, Pittsburgh, June 2-10, 2016

LANGUAGES

English, Bengali, Hindi, Marathi